

Start-Of-Season

1)

$$EVI2 = 2.5 * \frac{B8 - B4}{B8 + 2.4 * B4 + 1}$$

2)

$$LSWI = \frac{B8A - B11}{B8A + B11}$$

3)

$$2m_temp_max_t = \max[2m_temperature_t, 2m_temperature_{t-1}, 2m_temperature_{t-2}]$$

4)

$$skin_temp_max_t = \max[skin_temperature_t, skin_temperature_{t-1}, skin_temperature_{t-2}]$$

5)

$$GDD_2m_daily = \max[2m_temperature - 278.15, 0]$$

$$gdd_2m_acc = \sum_{n=1}^{\inf} GDD_2m_daily(n)$$

6)

$$DayLength_t = 24 - \frac{24}{\pi} \arccos(\tan(\varphi) \cdot \tan(\delta_t))$$
$$\delta_t = 0.409 \cdot \sin\left(\frac{2\pi}{365} DOY_t - 1.39\right)$$

Where φ is the latitude in radians, AKA:

$$\varphi = latitude * \frac{\pi}{180}$$

End-Of-Season

1)

$$EVI2 = 2.5 * \frac{B8 - B4}{B8 + 2.4 * B4 + 1}$$

2)

$$NDVI = \frac{B8 - B4}{B8 + B4}$$

$$NDVI3RE = \frac{B7 - B4}{B7 + B4}$$

$$RedContrast = NDVI3RE - NDVI$$

3)

$$2m_temp_max_t = \max[2m_temperature_t, 2m_temperature_{t-1}, 2m_temperature_{t-2}]$$

4)

$$skin_temp_max_t = \max[skin_temperature_t, skin_temperature_{t-1}, skin_temperature_{t-2}]$$

5)

Total Evaporation is taken from the dataset and needs not be computed manually.

6)

$$DayLength_t = 24 - \frac{24}{\pi} \arccos(\tan(\varphi) \cdot \tan(\delta_t))$$
$$\delta_t = 0.409 \cdot \sin\left(\frac{2\pi}{365} DOY_t - 1.39\right)$$

Where φ is the latitude in radians, AKA:

$$\varphi = latitude * \frac{\pi}{180}$$